

Cat Emotion Detection

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Abstract—Understanding animal emotions through visual cues can provide valuable insights into their welfare and behavior. In this work, I propose a hybrid deep learning approach and classical computer vision to detect and classify cat emotions from static images. Mysystem first uses a lightweight MobileNetV2-based binary classifier to detect the presence of a cat. Upon detection, a secondary MobileNetV2 model classifies the cat’s emotional state into one of seven categories: angry, upset, happy, normal, sad, scared, or surprised. To supplement the prediction, I introduced a tail position estimation module based on edge detection and contour analysis, which adjusts the final emotion confidence score. The combination of learned deep representations and heuristic tail cues improves robustness, especially in ambiguous cases. I demonstrated the effectiveness of my pipeline on a curated dataset and achieve promising results for practical applications in pet monitoring systems, veterinary care, and maybe animal behavior studies.

Index Terms—Cat detection, Emotion recognition, Animal behavior analysis, Deep learning, Computer vision, Tail position estimation, MobileNetV2, Edge detection

I. INTRODUCTION

It is important to understand how our pets feel, but one of the most challenging animals to interpret is the cat. Although dogs provide many clear vocal and physical signals when expressing emotions such as happiness, sadness, or anger, cats tend to be much more reserved. Although some cats may vocalize, much of their emotional state is conveyed through subtle body language, including posture, facial expression, and tail positioning. This subtlety makes automatic emotion recognition in cats a particularly difficult problem.

In this project, my aim is to bridge this gap by developing a computer vision system that detects the presence of a cat in an image and classifies its emotional state using deep learning models, supplemented by classical vision techniques for tail position estimation. Unlike traditional emotion classifiers that rely solely on facial features, my approach integrates both learned features and heuristic tail analysis to improve robustness and results.

The main contributions of this work are as follows:

- I proposed a two-stage pipeline combining a MobileNetV2-based cat detector and a MobileNetV2-based emotion classifier.
- I introduced a lightweight classical vision module that estimates tail orientation using Canny edge detection and contour analysis.
- I combined deep learning predictions with tail-based heuristics to refine emotion classification confidence.

Through experiments on a curated dataset, I demonstrated that incorporating body language features, such as tail position, minor improves emotion recognition performance.

II. RELATED WORKS

Prior research on automatic emotion recognition in animals has focused primarily on canine subjects, utilizing facial expression analysis to infer emotional states. For example, a recent study published in *Scientific Reports* [1] investigated the feasibility of using facial keypoints and convolutional neural networks to classify dog emotions. However, such studies remain limited in their ability to generalize between species due to anatomical and behavioral differences.

For feline analysis, progress has been comparatively slower. Existing efforts largely concentrate on facial recognition and tracking. For instance, Komarov [2] explored tracking a cat’s body using TensorFlow object detection models optimized for edge devices, demonstrating real-time monitoring capabilities. Similarly, work available through the NIH database [3] examined pet detection via facial features but did not extend toward emotional state classification.

These approaches share a common limitation: a predominant reliance on facial cues, which may not be sufficient for species like cats that communicate extensively through body language. In contrast, my method supplements emotion classification with an analysis of body posture, specifically tail orientation, to capture a broader range of emotional indicators. By integrating both learned features from deep learning and heuristic features from classical computer vision, my system tries to provide a more comprehensive framework for cat emotion detection.

III. METHODOLOGY

My approach consists of a multi-stage pipeline designed to first detect the presence of a cat, then estimate its emotional state using deep learning models, supplemented with classical computer vision techniques for tail analysis. This section details the datasets used, model architectures, training strategies, and post-processing techniques employed.

A. Cat Detection Model

For cat detection, I used the Oxford-IIIT Pet Dataset [4], a large collection of pet images covering 37 breeds of both cats and dogs. I filtered the dataset to include only images of cats,

discarding images of dogs. Each image was annotated with a binary label: *cat* (1) or *not-cat* (0).

I trained a MobileNetV2-based binary classifier using TensorFlow 2.x. The base MobileNetV2 model, pretrained on ImageNet, was used as a feature extractor with frozen weights to prevent overfitting given the moderate dataset size. I added a Global Average Pooling layer, a Dropout layer with a rate of 0.6 to prevent overfitting, and a final Dense layer with a sigmoid activation to output a binary classification probability. Initially I used a CNN model but quickly experienced overfitting due to the lack of layers used. I decided to use the pretrained model for convenience and accuracy.

The dataset was split into 70% training, 15% validation, and 15% testing sets. I applied standard data augmentation techniques including random horizontal flips, brightness shifts, and contrast adjustments. The model was trained using the Adam optimizer, binary cross-entropy loss, and early stopping based on validation loss with a patience of three epochs.

Upon convergence, the model achieved a validation accuracy of over 98% and was saved for inference as `cat_detector_mobilenetv2`.

B. Cat Emotion Recognition Model

For emotion recognition, I used the Cat Emotions Dataset available on Roboflow Universe [5]. This dataset categorizes cat images into emotional classes such as *Happy*, *Angry*, *Sad*, *Scared*, *Disgusted*, *Normal*, and *Surprised*.

I again adopted a MobileNetV2 architecture, freezing the convolutional base and fine-tuning a lightweight classification head consisting of a Global Average Pooling layer, Dropout (0.6), and a Dense softmax output layer with seven units corresponding to the emotion classes.

The dataset was divided into 70% training, 15% validation, and 15% testing splits using the `split_emotion_ds.py` script as originally the dataset only contained a training and validation set. Data augmentation was applied, including random rotations, zooming, and horizontal flips, to increase model robustness.

The model was compiled with categorical cross-entropy loss and the Adam optimizer. An early stopping strategy based on validation loss with a patience of three epochs was used to prevent overfitting. After training, the model achieved a validation accuracy of approximately 50-60% and was saved in, `cat_emotion_mobilenetv2`.

C. Tail Position Estimation

To incorporate additional non-facial emotional indicators, I developed a tail position estimation module using classical computer vision techniques. Given an input image, the following steps were applied:

- 1) Convert the image to grayscale.
- 2) Resize the image to a standard size (160x160 pixels) for consistency.
- 3) Apply Canny edge detection to extract prominent edges.
- 4) Extract contours using OpenCV's external contour retrieval mode.

- 5) Identify the second-largest contour as the likely tail contour (based on the heuristic that the largest contour corresponds to the cat's body).
- 6) Fit a line to the tail contour points using least-squares fitting.
- 7) Compute the orientation angle of the fitted line to categorize the tail as *Up*, *Down*, *Straight*, or *Curled*.

The estimated tail position, along with a confidence score proportional to the contour area, was then used to refine the emotion prediction.

D. Emotion and Tail Fusion Strategy

Recognizing that body language is an important signal of feline emotion, I designed a fusion strategy to combine the output of the deep learning emotion classifier with the tail estimation module.

Each emotion class was associated with expected tail positions based on observed feline behavior (e.g., *Happy* typically correlates with *Up* or *Straight* tails, *Sad* with *Down* or *Curled* tails). If the estimated tail position matched one of the expected positions for the predicted emotion, the softmax confidence score from the emotion classifier was boosted by a factor proportional to the tail detection confidence.

This post-processing adjustment helped resolve ambiguous cases where facial cues alone were insufficient, improving the robustness of the final emotion classification.

E. Inference Pipeline

At inference time, the following steps were performed sequentially:

- 1) Preprocess the input image (resize and normalize).
- 2) Apply the cat detection model. If the probability of cat presence exceeds a threshold (0.5), proceed; otherwise, classify as *No Cat Detected*.
- 3) Apply the emotion classifier to predict an initial emotional state.
- 4) Estimate tail position using the classical CV module.
- 5) Adjust the predicted emotion's confidence based on tail position matching.
- 6) Display the results overlaid on the input image using OpenCV.

IV. RESULTS

A. Cat Detection Model Performance

The cat detection model achieved excellent performance on the test set. Figure 1 shows the detailed classification report, including precision, recall, F1-score, and overall accuracy.

As evident from the metrics, the cat detection model achieved a very high accuracy, which is crucial for the overall system performance. Since the emotion detection step depends on correctly identifying a cat in the image, reliable cat detection forms a critical foundation for the pipeline. Misclassifications at this stage would propagate errors into downstream emotion analysis.

Classification Report:				
	precision	recall	f1-score	support
Not Cat	0.98	0.99	0.98	749
Cat	0.98	0.96	0.97	354
accuracy			0.98	1103
macro avg	0.98	0.97	0.98	1103
weighted avg	0.98	0.98	0.98	1103

Fig. 1: Cat detection model classification report showing accuracy, precision, recall, and F1-score.

B. Cat Emotion Detection Model Performance Without Tail Heuristics

Initially, I evaluated the emotion recognition model without incorporating tail position information. Figure 2 presents the classification report for this baseline setup.

Classification Report (Emotion Only):				
	precision	recall	f1-score	support
Angry	0.41	0.45	0.43	29
Disgusted	0.44	0.29	0.35	24
Happy	0.64	0.55	0.59	29
Normal	0.57	0.45	0.50	29
Sad	0.47	0.59	0.52	29
Scared	0.59	0.76	0.67	29
Surprised	0.59	0.59	0.59	29
accuracy			0.53	198
macro avg	0.53	0.52	0.52	198
weighted avg	0.53	0.53	0.52	198

Fig. 2: Cat emotion detection model classification report without tail position adjustment.

While the model performed reasonably well, achieving moderate precision and recall across most classes, the overall accuracy was lower compared to the cat detection model. This reflects the inherent difficulty of emotion recognition from subtle visual cues alone, especially in static images of cats.

C. Impact of Tail Position Heuristic

To improve emotion classification, I incorporated a tail position heuristic based on edge detection and contour analysis. Figure 3 illustrates the updated classification report after applying the tail-based confidence adjustment.

The inclusion of tail position estimation marginally improved classification performance. This suggests that tail posture provides useful auxiliary information when facial features or body language are ambiguous. However, the magnitude of improvement was limited, likely due to imperfections in the tail detection heuristic.

Specifically, the current method assumes that the second-largest contour corresponds to the tail, but in practice, the tail often merges into the cat's main body contour. As a result, the system's ability to accurately estimate tail orientation is constrained. Future work could explore more sophisticated

Classification Report:				
	precision	recall	f1-score	support
Angry	0.44	0.48	0.46	29
Disgusted	0.48	0.42	0.44	24
Happy	0.62	0.52	0.57	29
Normal	0.58	0.52	0.55	29
Sad	0.49	0.62	0.55	29
Scared	0.57	0.72	0.64	29
Surprised	0.67	0.48	0.56	29
accuracy			0.54	198
macro avg	0.55	0.54	0.54	198
weighted avg	0.55	0.54	0.54	198

Fig. 3: Cat emotion detection model classification report with tail position adjustment.

keypoint detection approaches or learning-based tail segmentation to address this limitation and better exploit body posture cues for emotion recognition.

V. DISCUSSION

The results reveal that determining an animal's emotional state, especially that of a cat, is a highly challenging task, even with modern computer vision techniques. Part of what makes this problem so difficult is that it mirrors the real world, even human observers often struggle to correctly interpret a cat's emotions without extensive experience or contextual information.

To truly achieve high accuracy in emotion classification, it would likely be necessary to collaborate with animal experts, such as veterinarians or behaviorists, to properly annotate training data. Expert knowledge could help define more reliable emotion categories and labeling guidelines, resulting in a stronger dataset.

Despite these challenges, the results of this project are encouraging. The models achieved reasonable accuracy considering the limited amount of available labeled data and the complexity of the task. Additionally, the incorporation of tail position, although heuristic and somewhat imperfect, provided a slight boost to emotion recognition performance. Future improvements in tail detection — perhaps moving beyond contour-based methods toward keypoint detection or segmentation could significantly enhance results.

Overall, these findings suggest that with continued refinement and a stronger annotated dataset, automated cat emotion recognition could become a feasible and valuable tool.

VI. CONCLUSION

While the project is still in its early stages, the potential applications and future directions are exciting. The next immediate step would be to create a larger, higher-quality dataset for emotion classification, with labels informed by veterinary expertise and comprehensive behavioral guidelines.

Throughout the development of this project, a broader vision began to emerge. In the future, this system could be expanded into an at-home veterinary AI application, where pet owners could submit videos or images of their cats. By

analyzing not only body language and tail posture, but also incorporating additional factors such as vocalizations, breed, age, and weight, the application could provide advice on a cat's emotional state and potential health issues.

Such a system could help reduce unnecessary veterinary visits for minor concerns, saving pet owners significant costs while ensuring cats receive the medical attention they truly need. Most importantly, it would contribute toward improving the quality of life for pets by giving owners better tools to understand and care for them.

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